

- 4 Collagen is a major component of the cartilage found in some of the structures of the human gas exchange system. Cells that synthesise and secrete the components of cartilage are known as chondrocytes.

(a) Fig. 4.1 is a transmission electron micrograph of a chondrocyte.

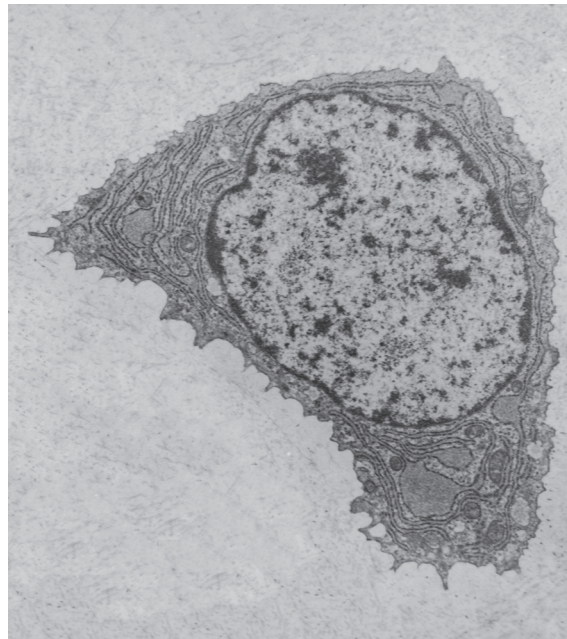


Fig. 4.1

With reference to Fig. 4.1, explain **two** features of the chondrocyte that show how the cell is adapted to its function.

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(b) (i) Describe the distribution of cartilage in the human gas exchange system.

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- (ii) Outline the function of cartilage in the human gas exchange system.

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- (c) Fig. 4.2 shows part of the primary structure of a collagen polypeptide.



Fig. 4.2

- (i) Name the type of covalent bond formed between the amino acids shown in Fig. 4.2.

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- (ii) Fig. 4.3 shows the molecular structure of the amino acid glycine (gly).

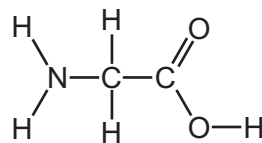


Fig. 4.3

With reference to Fig. 4.2 and Fig. 4.3 and the function of collagen, explain how the structure of a collagen polypeptide makes it suitable to form a collagen molecule.

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[Total: 10]